Specimen Handling and Transportation



Improper handling of specimens is not always detected before analysis. Therefore, all phlebotomists must follow the standard precautions guidelines when handling specimens. All phlebotomists must be instructed on policies and procedures for specimen handling. All specimens must be handled and transported carefully and delivered as quickly as possible to the laboratory for processing. Delaying the processing of specimens can affect metabolic changes in the sample. Harsh handling and agitation of a specimen can result in hemolyzed specimens, activation of platelets, and influence coagulation tests. Therefore, all tubes must be transported vertically and with stopper intake to reduce agitation and hemolysis. The upright position and intact stopper contribute and aid in clot formation and prevent clots from sticking to the stopper.

Blood specimens should be placed in a biohazard bag or container to be transported to the laboratories. The Clinical & Laboratory Standards Institute (CLSI) and OSHA require that specimens be marked with a biohazard logo, a liquid-tight closure, and a slip pocket for laboratory paperwork. For facilities where the laboratory is on-site, the specimen can be hand delivered by the phlebotomist or health care worker who collected the specimen to the laboratory or sent by automated internal transportation system.



Clinical laboratories receive patient specimens from locations outside of the facility. Many laboratories have courier services that periodically pick up specimens from physician's offices, clinics, patient service centers, or private homes throughout the day. Specimens should be properly stored; some may require refrigeration until courier services arrive. Criteria for the safe shipping of specimens include:

- The length of time designated for transporting the specimen
- The recommended temperature range required to maintain the specimen during transport
- Appropriate packaging of specimen during transport (i.e., protecting specimen from exposure to light)

Time Limits

Every specimen collected must be transported to the laboratory swiftly. According to CLSI guidelines, a maximum duration is recommended for a specimen to remain unprocessed. Immediate transport is feasible for on-site laboratories, such as those in hospitals. However, this quick turnaround is often not achievable for specimens collected at external sites. Typically, specimens from these locations are gathered and conveyed by courier services, which may exceed the advised 2-hr threshold. Consequently, facilities that collect specimens off-site should be equipped with processing areas where serum and plasma can undergo centrifugation, separation, and placement into a secondary vessel for transportation.

Temperature

Once blood is drawn from the body, its exposure to different temperatures can impact or compromise the analytes within. It is crucial to adhere to specific temperature guidelines to maintain the specimen's integrity during transit. The majority of specimens are stable for transport at ambient temperatures, approximately 25° C (77° F). However, if a specimen's temperature drops below normal body temperature, there is a risk of precipitation or agglutination occurring. Small, portable heat blocks can be used to transport a specimen that must remain at body temperature. These blocks are stored in a 37° C (98.6° F) incubator and can retain that temperature for around 15 min during transport. If a specimen can tolerate temperatures slightly above 37° C (98.6° F), wrapping it in an activated heat warmer is an option.

Conversely, ammonia levels in a specimen can increase quickly if left at room temperature. To ensure precise ammonia test results, such specimens must be immediately placed on ice or in a cooling tray, rushed to the laboratory (transported STAT), and processed to separate the plasma from the cells within 15 min after being drawn. Phlebotomists must be careful only to chill the tube containing the ammonia sample, as placing other tubes in contact with ice could lead to cellular changes that may cause the laboratory to reject the specimen. Rejected specimens necessitate a repeat blood draw for the patient, which is best avoided.

Specimen Temperature

Specimen temperatures related to specimen handling:

Body temperature: 36.4° C to 37.6° C (average 37° C); 96.8° F to 99.7° F (average 98.6° F)

Room temperature: 20° C to 30° C (25° C average); 68° F to 86° F (average 77° F)

Refrigerated temperature: 2° C to 10° C; 35.6° F to 50° F

Frozen temperature: -20° C or lower (some specimens require -70° F or lower; -4° F or lower (some specimens require -94° F or lower)

Photosensitivity

Substances being analyzed in blood can be broken down by light, resulting in false values. One test that is commonly impacted by light is bilirubin. A specimen for bilirubin loses 50% of the bilirubin after 1 hr of light exposure. One way to protect the blood and the substance within the blood is to wrap the tube in aluminum foil. Amber tubes, biohazard bags, and light-blocking transport containers can also transport blood to the laboratory.

Clinical Research

As a phlebotomist involved in clinical research, adhering to laboratory protocols is paramount to ensure the validity and integrity of the study results. Following these protocols meticulously is essential for maintaining the consistency and standardization required in a research setting.

Tube processing is the first critical step. Phlebotomists must be well-versed in the specific requirements for each type of specimen, including the correct order of draw, the type of anticoagulant or preservative in the tube, and the required volume of blood. Proper mixing and handling of the tubes post-collection are also crucial to prevent hemolysis or clotting, which could compromise the sample.

Obtaining the correct specimen bag is another important protocol. Specimen bags often have separate compartments for the specimen tubes and the associated paperwork, which must be filled out accurately and completely. This paperwork typically includes patient identification, time and date of collection, and specific study identifiers.

For samples that must be shipped, an air waybill is required. This document serves as a shipping label and a tracking form, which includes details about the sender, the recipient, and any special shipping instructions. Phlebotomists must ensure that the air waybill is filled out correctly to avoid any delays or mix-ups in transport.

Finally, packaging the specimens for transport must be done in compliance with all safety and regulatory guidelines. This often involves using biohazard bags, absorbent materials, and sturdy shipping containers that are designed to maintain the required temperature and protect the samples during transit. Phlebotomists must be familiar with the regulations for shipping biological materials, including international guidelines if the samples are being sent to a different country.